

Floor egg laying: can management investment prevent it?

D. L. M. Campbell ¹

Agriculture and Food, CSIRO, Armidale, NSW, Australia

Primary Audience: Researchers, Poultry Managers

SUMMARY

Floor egg laying can be a common problem for laying and breeder hens in cage-free housing systems. This is where hens lay eggs outside of the nest box, on the litter floor, slats, or tiered wire system. The causes of floor eggs are multifactorial which can make preventing floor egg laying behavior challenging and results in significant financial loss for egg producers. To better understand how floor eggs may be avoided, this literature review compiled information on factors that can affect the propensity of hens to lay floor eggs and management strategies that may prevent the undesirable behavior from developing. The review included studies on factors affecting nest box use, genetic factors, rearing and early training factors, production cycle factors, and new technologies being developed for floor egg laying detection and management. The information available online in producer-aimed management documents was also compiled and 10 egg producers within Australia were interviewed to understand current practices implemented within industry. Combined knowledge from multiple sources illustrated that floor eggs are manageable if the right strategies are implemented during the rearing period and when the birds first start laying their eggs. This is more challenging for egg producers that purchase point-of-lay birds and have limited control over rearing conditions. Understanding the behavior and needs of a particular hen strain or a particular flock is important given the multiple factors that can influence floor egg laying development. Floor egg laying may still develop if all causative factors are unable to be addressed.

Key words: poultry, laying hen, broiler breeder, system eggs, meat chicken, mislaid eggs

2023 J. Appl. Poult. Res. 32:100371

<https://doi.org/10.1016/j.japr.2023.100371>

INTRODUCTION

Floor egg laying, “mislaid eggs,” or “system eggs” by both domestic egg laying hens and (meat chicken and layer) breeder birds is still an industry problem. This undesirable behavior is where the eggs are not laid in the nest box in

cage-free systems and are laid on the litter floor, slatted floor, or tiered wire instead. This behavior is an issue because eggs laid outside the nest box results in food safety concerns with reduced saleability of contaminated eggs, broken eggs, increased labor for collection, and ultimately significant economic loss (van Horne, 1996; Jones et al., 2015). Floor egg laying is variable in its prevalence and often unpredictable in occurrence with variation across production systems, farms, flocks, and pens of the same flock (Mirosh et al., 1986; Appleby

© 2023 The Authors. Published by Elsevier Inc. on behalf of Poultry Science Association Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

¹Corresponding author: dana.campbell@csiro.au

et al., 1988a; van Horne, 1996; Abrahamsson and Tauson, 1998; Heerkens et al., 2015; Steinfeldt and Nielsen, 2015; Stratmann et al., 2016). However, floor egg laying is not a new problem. The last comprehensive literature review on this topic was published in 1984 (Appleby, 1984) and included stated reports of floor egg occurrence early in the 20th century (Pearl and Surface, 1909 as cited in Appleby, 1984) highlighting the longevity of this undesirable hen behavior. At the time of the Appleby (1984) publication, several factors were proposed to relate to floor egg laying occurrence although it was noted that “*there have been few properly conducted, statistically acceptable experiments on this subject*” (Appleby, 1984). Factors discussed to affect the prevalence of floor egg laying were multifactorial and included genetic effects (i.e., strain differences), design and management of the housing system, design and management of the nest boxes, bird training, and rearing influences (Appleby, 1984). It was concluded that although there may not be a multitude of systematic studies, when research findings were combined with commercial knowledge, factors contributing to floor egg laying had been substantially considered with the exception of how rearing may impact the behavior (Appleby, 1984).

The results presented in the Appleby (1984) review provided valuable guidance on what factors and/or management strategies may be successful in minimizing floor egg laying in cage-free systems. As a general overview, the review stated that genetic strain was indicated to affect both prelaying and perching behavior which can then affect access to and use of available nest boxes. When the hens are first moved to the laying facility, management measures should be implemented to train the birds to go to the nest boxes and prevent floor egg laying becoming established in the birds. This included disturbing hens sitting on the floor, immediately removing laid floor eggs, and destroying any nests found on the ground. Hens could also be picked up and placed into the nests or be temporarily confined within the nests. But some of these strategies may not be feasible in large flock sizes. Experience in perching/reaching elevated areas was indicated to be important for then accessing nest boxes in the laying facility. Thus, perches lacking during

rearing could negatively impact the perching behavior of the adult birds. As pullets typically do not have access to nest boxes during the rearing period, transfer to the laying facility must occur before the birds come into lay (Appleby, 1984).

Nest box accessibility and attractiveness were stated as key factors in whether hens will use them (Appleby, 1984). There must be sufficient nest boxes available per bird, travel distances to reach the nest box must not be large and access should not be obstructed by other parts of the housing structure. The nest box must also be more attractive/comfortable than alternative areas such as the floor. Specifics of optimal nest box design were difficult to draw conclusions on given studies assessing nest boxes often had variation in multiple aspects. Research results also varied on whether curtains were preferred or not, or whether hens do indeed prefer dark and secluded nest boxes or not, and different types of nesting substrates/materials or not (e.g., wire floor, mats, litter). Preferences may also depend on hen individuality, thus, a choice of nest box types could be a potential solution. Nest box use may increase if there are eggs already laid in there, including using decoy eggs, but there were no scientific studies on this management method. To deter birds from the floor, electric wires could block floor egg laying spots. There may also be impacts of lighting and ventilation where increased overhead lighting could reduce floor egg laying but limited systematic studies were available. Limiting feeding during peak laying times would reduce any conflict between eating and nesting behavior during oviposition (Appleby, 1984).

Almost 40 years following this review, floor egg laying remains a significant issue within the industry, but not for everyone. Some producers state they have solved the problem in their flocks with floor egg laying reduced to negligible proportions (see section on *Producer Interviews*). Given this is still an issue but may be manageable, the aim of this nonsystematic review is to summarize research on floor egg laying management in hens (domestic egg layers, their breeders, or broiler breeders) that has been published since the Appleby (1984) review. The terms “floor,” “mis-laid,” and

“system” eggs are used throughout the review depending on the relevance of the term to the research being cited as all terms are indicative of non-nest box laying. The review examines literature related to factors that affect floor egg laying including the nest box, genetics, rearing and training, production cycle management and technology development to aid in floor egg management in cage-free systems. In addition to the scientific literature, the management recommendations that are available online in producer-aimed documents have been compiled to draw on the knowledge that exists within the collective industry experience. Finally, the summarized outcomes of interviews with laying hen producers in Australia on management strategies they implement for preventing floor egg laying are included.

FACTORS AFFECTING FLOOR EGG LAYING

Nest Box Factors

Floor eggs can be viewed in different ways. This undesirable laying behavior can result from a lack of interest and motivation to access the nest box, active avoidance of the nest box because it is not an appealing laying site or an active preference to lay on the littered ground. Thus, floor eggs could be managed by both reducing litter attractiveness while increasing nest box attractiveness. The following section examines the evidence around factors affecting nest preferences in both laying hens and broiler breeders in cage-free and furnished cage systems. The literature around furnished cage systems is included as preferred features of nest boxes in this system may be relevant for similarly designed nest boxes in cage-free housing.

Evidence clearly demonstrates that nests are important to laying hens to fulfill their natural nesting behavior and that regardless of prior nesting experience, hens are highly motivated and willing to pay a price to access a nest site (reviewed in Weeks and Nicol, 2006; Cronin et al., 2012; Hemsworth and Edwards, 2021). Hens will exhibit prelaying behavior typically 1 to 2 h before they lay. This behavioral change comprises increased activity to search for a nest

site, sitting on the site and engaging in nest building behavior (reviewed in Cooper and Albentosa, 2003; Hemsworth and Edwards, 2021). While domestic hens have predominantly lost their broodiness, they will still exhibit this prelay behavior and generally prefer discrete, enclosed spaces in which to lay their eggs (Weeks and Nicol, 2006). Denying nest access will result in behaviors indicative of frustration although there is limited supporting evidence that a physiological stress response is induced in the absence of a nest box (reviewed in Hemsworth and Edwards, 2021). Hens may also exhibit frustration behaviors if they are unable to locate a suitable nesting site during the prelay phase. In an experimental free-range setting, hens were kept in litter floor pens without nest access until 21 wk of age following which they were given pasture access with potential natural vegetation nest sites (Kruschwitz et al., 2008b). The birds were already laying at this age and exhibited frustration behaviors when laying in the pasture compared with those hens that had access to an enclosure in the pasture area containing a familiar litter laying substrate (Kruschwitz et al., 2008b). Similarly, Freire et al. (1996) observed some components of prelaying behavior to increase when hens were presented with a more unfavorable nest option. Nest type and hen preferences can affect the eventual choice of the nest for laying their egg, and/or their exhibition of prelay behavior (Struelens et al., 2008; Lentfer et al., 2011).

Not all hens will lay inside a nest box initially, indicating there is an experiential component for increased nest box use across age; however, some hens will never lay in a nest box (Appleby et al., 1988a,b; Sherwin and Nicol, 1993a; Cooper and Appleby, 1996; Bestman and Wagenaar, 2014; Oliveira et al., 2019). There are differences observed in prelaying behavior between nest box and floor layers suggesting there may be inherently different types of laying hens with consistent preferences for litter or a nest box, different perceptions of a nest box, and/or possibly avoidance of a nest box (Rietveld-Piepers et al., 1985; Appleby et al., 1986; Cooper and Appleby, 1996, 1997; Cronin et al., 2005; Kruschwitz et al., 2008a; Zupan et al., 2008). Behavioral differences in

prelay behavior have also been observed between hens that lay in the nest or scratch area in furnished cages (Hunniford and Widowski, 2016). Hens that consistently preferred to lay in a litter tray vs. a nest box in individual floor pens exerted the same amount of effort against weighted doors to access the litter tray as those that accessed the nest box (Kruschwitz et al., 2008a).

As distinct behavioral types, nest and floor layers are typically consistent in their relative location preferences. Individual hens may also be consistent in their preferences for a particular laying location and specific nest box features. In both commercial and small-scale experimental settings, floor-housed birds were observed to show some, but limited consistency in repeated use of a specific nest box by an individual hen (Rietveld-Piepers et al., 1985; Appleby and McRae, 1986; Appleby et al., 1986). Other research has found high consistency in choice of a particular nest box when there was a choice of 2 nest boxes in a furnished cage (Wall et al., 2004). Individual broiler breeder hens while not consistent in their use of a particular nest box, were consistent in their choice of features of a nest box when there was heterogeneity in what was available (e.g., height of nest box from the ground; Appleby et al., 1986). When single and triple nest boxes were moved to different positions in the floor pen, individual hens varied in whether they kept picking the same laying location, or same degree of laying isolation (Riber and Nielsen, 2013). When given a choice between a preferred nest height (i.e., ground level) or a nest in a familiar location, hens picked the familiar location (Krause and Schrader, 2018). This highlights how certain features may be preferred over others and can depend on the individual hen. Thus, similar to research showing that hens vary in their use of different areas within a housing system (e.g., inside/outside in a free-range system: Campbell et al., 2020; tiers and litter area of an aviary system: Campbell et al., 2016) variation in nest box choice options within a commercial system may better cater to all hens (Petherick et al., 1993; Sherwin and Nicol, 1993b).

Given the laying preferences that individual hens develop, once this preference is

established, it may be difficult to modify it and/or there will be a period of significant disruption while hens adapt to a new nest. Individual hens in wire-floor cages with a laying preference for 10 wk did not modify that preference when a preferable artificial turf substrate was added to a different part of the cage (Hughes, 1993). In commercial aviaries, moving the nests from being integrated to against the wall and vice versa resulted in almost all eggs being laid on the litter following the change (Lentfer et al., 2008). The hens showed behaviors indicative of agitation/frustration although there was eventual adaptation to the change (Lentfer et al., 2008). Sherwin and Nicol (1994) showed some hens continued laying in the same location regardless of changes in floor substrate type in individual cages whereas others changed location to lay on a preferred flooring type. Similarly, Riber and Nielsen (2013) showed hens changed their preferred nest box when the artificial turf mat was removed leaving only wire flooring.

If hens are able to establish a preference for the nest box during the early laying period and the nest box features/microclimate remain consistent, then the preference should be set for the flock cycle. Thus, it is important to ensure the nest box available in the laying system is going to be perceived positively and utilized well (Buchwalder and Fröhlich, 2011). Some commercial nest box designs have resulted in higher nest box smothers than others indicating that if the design is not optimal, there can be other negative behavioral consequences besides floor egg laying (Rayner et al., 2016). There are multiple nest box design factors which have been shown to affect prelaying behavior, nest choice and preferences of laying hens and broiler breeder hens. For example, the color of curtains, the nest box, and nest pad (Brake, 1993; Huber-Eicher, 2004; Clausen and Riber, 2012; Pillan et al., 2023), which may even be influenced by the color and light intensity experiences of the chicks (Huber-Eicher, 2004; Zupan et al., 2007). Preferences for the slope of the nest floor (Stämpfli et al., 2011), presence of curtains (Struelens et al., 2005; Struelens et al., 2008; Hunniford and Widowski, 2018; Sun et al., 2023); curtain design (single or multiple pieces) (Stämpfli et al., 2012), construction

material of the nest (Holcman et al., 2007; van den Oever et al., 2020a), nest size (Ringgenberg et al., 2014; Ringgenberg et al., 2015a), nest height (Krause and Schrader, 2018), and nesting/floor substrate type and/or depth (Appleby et al., 1988c; Petherick et al., 1993; Wall et al., 2002; Struelens et al., 2005; Struelens et al., 2008; Guinebreti re et al., 2012) have also been investigated. Preferences may be relative to what is available, and some features may be weighted over others. Hens will show preferences for artificial turf over wire (Hughes, 1993; Struelens et al., 2005) but manipulable substrates over fixed flooring types when given that choice (Brake, 1985; Huber et al., 1985). Hens prefer a manipulable substrate in the nest, where an open litter box was the most preferred nest choice when compared with a nest with roof and 3 walls and a thin plastic mat on the floor (Buchwalder and Fr hlich, 2011). However, when a curtain was added to the front of this nest, the hens then preferred it over the open litter box indicating a fully enclosed space may be more important than the laying substrate (Buchwalder and Fr hlich, 2011). Similarly, hens in furnished cages showed preferences for different flooring types, dependent on whether the nest was also enclosed or open (Hunniford et al., 2018). The enclosed space was more valued than substrate type (Hunniford et al., 2018). There may also be factors beyond the nest itself that affect how it is perceived and used in a commercial system. This can include where the nests are placed—either integrated into the tiered aviary unit or separate against the opposite wall, the width and material of the access platform in front of the nests (Lentfer et al., 2011; St mpfli et al., 2013), and the placement of drinker lines in relation to the nests (Lentfer et al., 2013).

An individual hen may have preferences for a nesting site, but given hens are housed with hundreds or thousands of other individuals there are social factors that will also be influencing where a hen will lay. Rietveld-Piepers et al. (1985) showed no effect of social dominance status on floor or nest laying. However, other research has suggested subordinate hens may be excluded from nest access leaving only the floor as a laying option (Holcman et al., 2007). There can be competition for the more

preferred/popular nesting sites and pecking aggression may affect nesting behavior. Observations of nesting in nest boxes and corners of experimental floor pens with different (small) flock sizes showed corners were preferred laying spots despite there being greater competition for these areas in the larger flock sizes including more aggressive pecks (Lundberg and Keeling, 1999). However, if a bird was getting pecked at, she stayed at the site for a shorter amount of time, and those birds delivering the pecks stayed in the corners for longer (Lundberg and Keeling, 1999). These observations highlight the impact of social variables on the time that a hen may spend at a particular nesting site (Lundberg and Keeling, 1999). In an experimental testing scenario, hens were less willing to enter a corridor to access a nest box when there was an unfamiliar or dominant bird present relative to a subordinate or no conspecific and they tried to find alternative routes to access the nest (Freire et al., 1997).

A key social factor beyond dominance ranking affecting laying hen nesting in commercial systems is gregarious nesting where a hen is more likely to occupy an already-occupied nest (Riber, 2010). This may increase the risk of smothering in the nest, heat stress, scratch/peck wounds, as well as affect egg quality (more broken and dirty eggs) and increase floor egg laying (Riber, 2010; van den Oever et al., 2021a). The propensity for this behavior in domestic hens varies between individuals (Sherwin and Nicol, 1993a; Tahamtani et al., 2018). It also contrasts with isolated nesting seen in the hens' wild counterparts (Riber, 2012a). Domestic strains allowed free choice for varying nest sites (natural or artificial opportunities presented via vegetation and plywood sheets) in a semi-natural setting still showed gregarious nesting behavior (Riber, 2012b). This may, in part, have been affected by nest site options. Gregariousness was only observed in nests along enclosure borders which were presumably the more preferred nesting sites vs. those nesting sites 1 m+ away from the border (Riber, 2012b). Gregarious nesting may be derived from an antipredator response to nest predation (Riber, 2012a). Although research with small groups of hens found no relationship between fearfulness (nor sociality, or dominance) and a hen's tendency to be solitary or gregarious

nesters (or a mix between both) (Tahamtani et al., 2018).

Gregarious nesting may be affected by the age of hens and their experience in selecting their own nest site where inexperienced hens may be more likely to visit already occupied nest sites (Riber, 2010). This may, in turn, contribute to the higher use of particular nest boxes within systems which are typically corner or end nest boxes or particular sides of nest boxes (Riber, 2010; Clausen and Riber, 2012; Riber and Nielsen, 2013; Ringgenberg et al., 2015b; Villanueva et al., 2017; Li et al., 2018). Alternatively, a specific attractive nest box can increase the observed gregarious nesting (Clausen and Riber, 2012; Giersberg et al., 2019). Broiler breeder hens have been shown to crowd in highly preferred nests resulting in greater aggression (van den Oever et al., 2020a). If a system is to provide heterogeneity in nest boxes to try and minimize crowding in specific boxes, then it is suggested that modifications to less preferred middle nest locations may improve nest occupancy uniformity (Clausen and Riber, 2012). However, trials with variation in nest curtain color and identifying symbols did not affect nest visits in experimental floor pens relative to other factors such as nest position and occupancy (Ringgenberg et al., 2015b). To ensure sufficient nesting space for hens in commercial cage-free systems, colony nest boxes are often large. This may decrease the hens' preference for the nest box. Testing with smaller numbers of hens in floor pens showed a preference to lay more eggs in smaller sized nests (i.e., larger colony nests with dividers) compared with the more open space of the typical commercial colony nest (Ringgenberg et al., 2014; Ringgenberg et al., 2015a; Zheng et al., 2018). Ensuring the nest box is optimally attractive is challenging as there are multiple design-related and social factors to consider.

Genetic Factors

The original floor egg laying review indicated behavioral differences between hen strains which contributed to the occurrence of floor eggs (Appleby, 1984). This effect of genetic line, particularly brown vs. white hens has been evidenced across several further

studies but with contradiction in whether the research finds that white or brown hens tend to lay more floor or system eggs (Campo et al., 2007; Singh et al., 2009; Heerkens et al., 2016; Villanueva et al., 2017; Purdum et al., 2020; Rizzi, 2020; Farkas et al., 2022 but see van den Oever et al., 2020b for no differences in floor egg laying across different broiler breeder genetic lines). In alignment with these differences in laying behavior, there is evidence of the variation between white and brown hens in how they use a tiered aviary system, including the nest box (Ali et al., 2016; Villanueva et al., 2017; Ali et al., 2019a; Ali et al., 2020; Purdum et al., 2020). This in turn, has an impact on whether eggs are laid in the nest box, the wire flooring within the system, or on the litter (Villanueva et al., 2017; Purdum et al., 2020). Brown strains are often heavier than white strains which may contribute to reduced vertical movement and lower use of perches (Silver-sides et al., 2012; Purdum et al., 2020). If hens are less motivated to access the upper tier where the nest box is located or spend a greater amount of time on the litter, there is a greater chance of eggs being laid outside the nest box (Ali et al., 2020; Purdum et al., 2020).

White and brown strains have been shown to vary in oviposition timing (Campo et al., 2007; Icken et al., 2012; Villanueva et al., 2017; Becot et al., 2021). Some research indicates white strains compared to brown strains will lay most of their eggs within a 3-h window (Icken et al., 2012), whereas other research indicates the majority of eggs from brown strains are laid within a 4-h window (Villanueva et al., 2017). Brown strains do appear to lay their eggs earlier than white strains (Lillpers, 1991; Campo et al., 2007; Icken et al., 2012; Villanueva et al., 2017; Becot et al., 2021) and brown or white strains vary in duration spent occupying the nest box (Icken et al., 2012). Moderate to high heritabilities (Lillpers, 1991) or low to moderate heritabilities (Icken et al., 2013) have been quantified for time of oviposition and low to moderate heritabilities for duration of time occupying the nest box (Icken et al., 2013). These genetically predetermined oviposition times could result in competition for nest box resources at certain times of day contributing to more eggs being mislaid

(Villanueva et al., 2017; Ali et al., 2020) and thus selection should maintain individual differences (Becot et al., 2021). Variation among different strains suggests some strains may perform better in certain types of housing systems or housing system designs (Tauson et al., 1999). System resources should be modified to accommodate strain behavioral variation (Villanueva et al., 2017; Purdum et al., 2020) and calculation of sufficient nest box space needs to consider hen preferences for nest boxes on certain levels only (Farkas et al., 2022).

In addition to differences across genetic lines, there is also evidence that variation in nest preferences, time spent in the nest and floor egg laying behavior/nest box acceptance within white and/or brown strains is heritable (Settar et al., 2006; Becot et al., 2021; Wolc et al., 2021; Bécot et al., 2023) indicating selection against undesirable traits may be possible. Selection for higher numbers of eggs laid in a trap nest across 5 generations with different hen lines did see a reduction in the percentage of floor eggs (Sørensen et al., 2017). In the breeding environment at the level of genetic selection, those hens that do not produce a higher number of eggs in the nest are selected against; however, the selection is not suggested to be an all-encompassing solution (Preisinger, 2018). It may be that there is a genetic component for the ability to learn nest box use which, in part, could be related to perching ability (Wolc et al., 2021). This suggestion of the relevance of learning ability highlights the importance of adequate rearing and training when the birds are developing and coming into lay, respectively, to prevent floor egg laying (Preisinger, 2018; and see following section *Rearing and early training related factors*).

Rearing and Early Training-Related Factors

The Appleby (1984) floor egg laying review highlighted that at the time, there was comparatively more limited consideration of rearing influences in the management of floor egg laying. To address this gap in knowledge, Appleby et al. (1988b) conducted a study to clearly demonstrate the benefits of perch provision during rearing for reducing floor egg laying in adult birds, recommending early placement of

perches (by at least 4 wk of age). This was further supported by Gunnarsson et al. (1999) who looked at risk factors for floor egg laying across 59 flocks of cage-free birds on 21 different farms and found the only significant correlating factor to reduce percentages of floor eggs was access to perches no later than 4 wk of age. Although the authors did highlight that rearing with perch access was not a complete solution to floor egg laying and additional factors would still need to be managed (Gunnarsson et al., 1999). Similar effects have been seen in broiler breeder birds where rearing (from day-old) with perches similar to the nest box perches they would encounter during lay, significantly reduced floor egg laying when the breeder birds were moved to a slat-litter laying house (Brake, 1987). Even provision of perches from 8 wk of age still reduced floor egg laying in breeder flocks during the peak lay period (Appleby et al., 1988b).

There is now a much greater understanding of the short- and long-term impacts of rearing environments on how pullets develop physically, physiologically, and behaviorally (Janczak and Riber, 2015; Campbell et al., 2019). Generally, there is a recommendation to rear birds in the same type of systems they will be housed in during lay to optimize their adaptation and performance following transfer to the laying facility (Abrahamsson and Tauson, 1998; Colson et al., 2008; Janczak and Riber, 2015). The like-to-like benefits were demonstrated by Colson et al. (2008) who looked at behavior and adaptation to (i.e., use of) an aviary system following rearing on the floor (with perches and platforms), or aviary rearing. After transfer to the laying facility at 17 wk of age, those birds that were floor-reared showed poorer use of the aviary upper levels with a preference for the litter and lower levels. This preference resulted in fewer eggs laid in the nest box and more laid on the litter across the early-to-peak lay period relative to the birds that had been aviary-reared (Colson et al., 2008). Further evidence for perch access benefits comes from research looking at the effects of transferring pullets to an aviary at 17 wk of age, or providing them perches and nests in floor pens at 17 wk of age, or keeping them in floor rearing pens without elevated structures

(Ali et al., 2019b; Karcher et al., 2019; MacLachlan et al., 2020). This situation would simulate if there was a requirement to delay pullet transfer such as during a disease outbreak. With these treatments in place until 25 wk of age when all the birds were moved into experimental aviaries, those pullets given perches and nest box access laid fewer floor eggs than the empty floor pen treatment (Karcher et al., 2019; MacLachlan et al., 2020). This was likely related to the reduced use of the upper levels of the tiered system by floor-housed birds (Ali et al., 2019b). The differences between treatments had lessened by 54 wk of age showing the birds could adapt at a later age, but there were still more floor eggs laid by those birds that had no nest box and vertical structure access during the early laying period (MacLachlan et al., 2020). Similarly, an experimental study where birds were transferred at 22 wk from conventional cage rearing to an aviary system showed over 95% floor egg laying across the 10-wk trial period (Philippe et al., 2020). Other beneficial rearing effects include the use of dark brooders for layer chicks which mimics the natural brooding environment that a mother hen would provide. Chicks reared with dark brooders in an experimental setting showed reduced floor egg laying during the early-to-peak lay period compared to control birds reared in standard open floor litter conditions (Riber and Guzman, 2017).

The rearing period before transfer to the laying facility is important for pullets to develop movement skills and familiarity with a system type, but the initial period after transfer to the laying facility is also critical. The exact age at when pullets are transferred varies and can depend on multiple factors such as production shed schedules, pullet vaccination schedules, labor, and the cost of feeding birds not producing eggs. The typical transfer age is around 15 to 17 wk. Egg laying then begins around 18+ wk of age, but hens will start looking for a suitable nest site prior to their first egg (Rietveld-Piepers et al., 1985). This highlights the importance of familiarity with nest box laying locations before oviposition onset to reduce egg laying outside of the nest (Sherwin and Nicol, 1993a). There is evidence across multiple studies that nesting behavior has a learned

component where the percentages of floor eggs are typically highest when the hens first start laying and reduce across time (Cooper and Appleby, 1996; Settler et al. 2006; Bestman and Wagenaar, 2014; Oliveira et al., 2019; Wolc et al., 2021). Thus, varying strategies of training will occur when the birds first move into their production housing. For example, Pillan et al. (2023) showed that the use of nest lights coming on 1.5 h before the main house lights increased nest box use from 17 to 26 wk compared to hens using nests without the low intensity lights in an experimental aviary system.

Other management strategies following pullet transfer can include restricting access to litter until the birds have familiarized themselves with the cage portion of the system (i.e., food, water, nest boxes, perches, tiers), litter restriction until the hens have reached peak lay, or continued litter restriction each morning across the lay cycle until the majority of daily eggs have been laid (Nicol et al., 2003; Zhao et al., 2015; Ali et al., 2016; Oliveira et al., 2019). Outdoor access in free-range systems is also typically restricted after pullet transfer until laying commences and until after the morning laying period across the production cycle (Rana et al., 2022). However, early work by Mirosh et al. (1986) found exposure to a range area during onset of lay actually decreased floor egg laying, possibly because the pullets were more motivated to seek out secluded nest areas compared to pullets kept in an enclosed system. Experimental evidence for the impacts of litter restriction is limited. Testing of a 2-wk litter restriction period upon first transfer to the laying facility showed no impact on percentage of floor eggs across the laying cycle (Alm et al., 2015), but it is unclear if this was compounded by the nest boxes being closed for the first 12 d following transfer until the first egg was laid (Alm et al., 2015). In contrast, partial restriction of litter access each morning across the laying cycle resulted in a significant reduction in floor egg laying compared with hens that had continuous litter access (Oliveira et al., 2019). The aviary-reared pullets were transferred to the laying facility at 17 wk of age and all birds were restricted to the tiered portion of the system for 10 d to learn the placement of resources, including the nest box (Oliveira et al.,

2019). The hens that were given partial litter access had a total of 4 wk before litter access was allowed, compared with only 10 d restriction for the comparative continual litter access treatment groups (Oliveira et al., 2019). It is likely the initial longer period of litter restriction trained the birds to lay in the nests with this reinforced by the continual partial restriction across the flock cycle. Although litter restriction is implemented by producers to avoid a floor egg problem (Palczynski et al., 2016), the restriction can be a risk factor for feather pecking (Nicol et al., 2003; Lambton et al., 2010), highlighting the management balance that producers have to work by to mitigate multiple undesirable hen behaviors (Palczynski et al., 2016).

Production Cycle Factors

While there are different management strategies that can be applied during rearing and early lay to prevent floor egg laying, it is generally understood that if floor egg laying is established in the flock, there are few options for remedying this. However, there may be factors of the adult hen laying system housing design that contribute to floor egg percentages. For example, increased floor egg laying may be a consequence of the system type; free-range compared with deep litter (Campo et al., 2007), or of the structural housing system design, such as the specific type of aviary system in place (Stratmann et al., 2016). Housing design modifications may also impact floor egg laying. In experimental testing conditions, broiler breeder birds reared with perches and platforms were given access to floor pens that included a raised slatted area in front of the nest box or not and the slatted area decreased floor eggs (van den Oever et al., 2021b). The drinker nipples were also up on the slatted area which may have encouraged the birds to go up to the nest boxes (van den Oever et al., 2021b). Comparatively, in the treatment with no slats, the drinker lines and nest boxes were in the litter area (van den Oever et al., 2021b). In contrast, Stratmann et al. (2015), found no effect of additional ramps, perches, or platforms in an aviary system for aviary-reared birds on the percentages of floor eggs. Modifications of aviary design to

facilitate greater movement around the aviary (inclusion of ramps and/or removal of internal partitioning panels) also did not affect the percentages of system eggs (Nannoni et al., 2022). However, it was observed in one of the treatment modifications that the highest concentration of eggs laid on a particular aviary tier did equate to the highest proportion of system eggs, suggesting possible nest space competition (Nannoni et al., 2022). Increased stocking density can increase floor egg percentages, likely also a result of greater nest space competition (Kang et al., 2016). Perches provided during the laying period in a free-range system for birds that had been reared with perches had no impact on the percentage of floor eggs relative to perch-reared birds without specific aerial perch access during lay (birds still perched on water lines and feeders) (Donaldson and O'Connell, 2012). As discussed previously (see section on *Genetic factors*) floor egg laying may be a consequence of the strain of hen being housed and an interaction of housing design with strain behavior. Heerkens et al. (2016) found inconsistent strain-dependent effects of additional ramps in experimental aviary-style systems for reducing floor egg laying.

There are a small number of research studies that have looked at manipulating factors across the laying period to either directly or indirectly look at the impacts on floor egg laying. Inclusion of 1.5% of experienced hens in an aviary system during the laying period did not affect the percentages of floor eggs, but it was not clear from the study if this was a result of the (low) density of experienced hens or that they did not have influence on the younger (aviary-reared) hens' behavior (Oliveira et al., 2019). Environmental enrichment of hay provided across a flock cycle in aviary systems significantly reduced the percentage of eggs laid outside the nest box although the numerical difference was small as mean mislaying percentages overall were only around 1% (Son et al., 2022). In contrast, Schreiter et al. (2020) found the provision of pecking stones and alfalfa bales during both rearing and lay increased the percentage of floor eggs which the authors suggested may have, in part, been due to increased time spent on the litter area using the enrichments (time on litter was not explicitly

measured). Finally, [van den Oever \(2020b\)](#) looked at health-related factors across 5 broiler breeder genetic lines but found no relationship between hock and footpad contact dermatitis and floor egg laying indicating the behavior was not a direct result of mobility differences. Overall, research across the laying period is limited as the key periods for behavioral manipulation are when the birds are younger.

TECHNOLOGY ADVANCEMENTS

With the increase in precision technology for agricultural management, there are considerations of how advanced technologies could help prevent and/or manage floor eggs ([Ren et al., 2020](#); [Subedi et al., 2023](#)). Floor robots are commercially available (e.g., Spoutnic, T-Moov) to encourage hens to move off the floor rather than establish litter nests and floor egg laying behavior. This may help to train the hens to use the nest boxes initially, replacing or complementing frequent house walk-throughs during this period. However, robust evidence of their effectiveness in the literature is currently limited ([Li et al., 2022](#)). There may be some effect to reduce floor egg laying, but hens do rapidly acclimate to the robot and can be observed riding on it after a couple of weeks ([Li et al., 2022](#)). Several different robots for floor egg detection and collection have also been the subject of precision technology development ([Ren et al., 2020](#)). These have shown varying degrees of egg harvesting success for potential reduction in the labor requirements when there is a floor egg issue across the production cycle ([Vroegindeweij et al., 2014](#); [Joffe and Usher, 2017](#); [Usher et al., 2017](#); [Vroegindeweij et al., 2018](#); [Li et al., 2021](#)). These robots contain camera-based automated algorithms for egg detection with varying egg pickup techniques (e.g., suction cup, gripper, helical spring), and movement trajectory algorithms spanning the entire floor area including frequent visits to high-risk areas ([Vroegindeweij et al., 2014](#); [Joffe and Usher, 2017](#); [Li et al., 2021](#)). Sensors within the robots enable navigation among birds and other obstacles in the complexity of cage-free environments ([Usher et al., 2017](#); [Vroegindeweij et al., 2018](#); [Chang et al., 2020](#)). There is still research

into further refining and validating detection algorithms ([Li et al., 2020](#)).

Ongoing research includes the possibility of automatically detecting hens sitting on the ground (particularly in corners) indicative of floor egg laying behavior ([Bist et al., 2023](#)) and developing fixed camera detection systems that can automatically monitor floor eggs within the house using machine vision ([Subedi et al., 2023](#)). It is anticipated that the automated technologies could be valuable as a training tool when production commences to remove the eggs and not attract other hens to floor lay, as well as across the production cycle if the floor egg laying persists to reduce labor requirements. Ultimately, radio-frequency identification-enabled smart nest boxes that can track individual hens' laying behavior may allow producers to detect poorly performing birds, at the very least, improvements in nest box technology facilitates genetic selection and research at the individual hen level ([Chien and Chen, 2018](#)).

PRODUCER INTERVIEWS

To collate the knowledge of industry on floor egg laying management that is not captured in the scientific literature, semi-structured interviews were conducted with 10 egg laying hen producers in Australia (approval via CSIRO's Social Science Human Research Ethics Committee in accordance with the National Statement on Ethical Conduct in Human Research 2007 (Updated 2018): Approval number 015/23). Thirty-five producers (or former producers now working as technical poultry consultants out of a total of 85 egg production businesses covering 80% of the national flock) were initially directly contacted (phone call, email, or website contact forms) and invited to participate by providing a short summary of the study aims, the topic areas of the questions, expected time commitment and that all answers would be compiled anonymously following participation. Direct contact was chosen as more likely to result in participants vs. a general invitation to all producers. A total of 10 producers (8 primary producers and 2 technical poultry consultants) agreed to be

interviewed and voice-recorded and were provided the human ethics participant information sheet prior to the interview. A series of 10 questions (see [Supplementary Table 1](#) for the list of questions) were asked in a qualitative conversation style commencing with questions about years of experience in hen management, type of housing system they work with, hen strain they house, typical flock size, and type of nest boxes. The producers were also asked whether they rear their own pullets or if they buy them in, and if yes to the latter, how much knowledge and control do they have over the rearing process. They were then asked if they currently have a floor egg problem, if it varies by flock, and the extent of their problem (i.e., percentage estimate of floor eggs). Whether the answer to this question was “yes,” or “no,” all producers were asked what management strategies they implement or have tried to implement to reduce their floor egg problem. This question resulted in the lengthiest answers depending on the depth the interviewee went into with back-and-forth exchange between the interviewer and interviewee allowing the conversation to evolve. Interviewees were asked whether these management strategies were based on trial-and-error, reading literature/online documents and management guides or talking with other producers. Finally, the interview ended with an open-ended question giving the producers a chance to provide any other information they felt was missed by the previous questions. The interviews were voice-recorded and later transcribed for accuracy of data capture with all information gathered by a single researcher (DLMC).

Across the 10 producers, the years of experience managing laying hens ranged from 5 to 40 with a combined total of 162 years (median 8.5 years). All producers worked with brown strains, but some had experience working with other strain types previously. Every producer currently worked with free-range systems, but some also worked with (or had worked with), aviary, barn, and slatted systems. Flock sizes were varied across all producers ranging from 3000 up to 50,000 birds. Three producers bought their pullets in from contract growers at 16 wk with no control over how they were reared (these were producers from smaller-sized farms) with the remaining producers

either rearing their own or contracting the rearing but with knowledge and control over the rearing process. Nest boxes were all colony and curtained that came with the installed systems. Of the 10 producers, 4 stated they were not concerned about floor eggs as their management strategies kept mislaid eggs to a negligible number, the remaining producers indicated they did have a problem, ranging from around one to 15%, with inconsistent variation across flocks. All producers indicated their strategies to (try and) manage floor eggs were typically based on trial-and-error with 4 producers stating they talked to other producers, 4 producers read advice online in breeder manuals/housing system manuals, one producer talked with a poultry consultancy company and 2 producers read scientific literature.

The management strategies that producers implemented were relatively consistent across all interviewees but were dependent on the labor/financial resources available to producers and the extent of control they had during the pullet rearing process. The strategies centered around rearing and early lay training as critical time points for modifying bird behavior. Generally, the key aspects were making sure you were rearing like-to-like, that is, if birds were going to be placed in an aviary system at lay, they needed to be reared in an aviary system so they would know how to go up (to the nest box) at lay. This was feasible for those producers that reared their own birds but less so for producers that were only able to buy in pullets at 16 wk. These producers expressed frustration at their lack of control over the rearing process. Furthermore, there was uncertainty around what condition the pullets may arrive in. For example, if there were any issues during the rearing process that would affect the bird behavior, growth, and adaptability, this was unknown to the receiver at time of arrival. For those producers that had control over the rearing process, they transferred pullets over to the laying shed earlier than 16 wk (14–15 wk) to give them more time to adapt to the laying housing and become familiar with the set-up, including nest box location, prior to first oviposition. While an early transfer (i.e., early buy in of pullets) may partially mitigate lack of control over rearing for smaller producers, it was not

economical to do so given the cost of feeding birds that are not yet laying and/or lack of empty sheds not occupied by hens still laying.

Ramps were used in the laying shed initially by 3 producers to improve the transition between rearing and laying housing. Another factor that was stated by every producer interviewed was the importance of floor/shed walks at the beginning of the laying period to keep the birds moving up off the floor into the nest boxes, to pick up eggs laid on the ground, and to ensure that the nest box was the only quiet place for the birds to lay their eggs. Producers stated it was not just about doing regular floor walks, but these walks needed to be done by competent staff who knew what the purpose of the walks were and were willing to invest the time and effort to do them properly. Every producer recognized this management investment for the first approximate 6 wk was critical to establishing good laying behavior in their birds for the remaining flock cycle. However, 3 of the smaller producers stated they either could not afford this labor investment as it was more than the financial loss from unsaleable floor eggs, or they could not find enough good staff to complete this job, so they accepted a certain percentage of floor eggs were inevitable. One producer said they were told larger-scale producers may accept the loss of floor eggs as they could more easily afford to. There was one producer that looked for broody birds during the floor walks and these were culled.

Producers all varied in how they managed the nest boxes, whether they were closed when the birds arrived in the shed to prevent roosting and/or smothering in them, whether they were open to allow the hens to start becoming familiar with them before laying commenced, or whether half were open and half were closed to encourage a more even distribution of nesting. Lighting was mentioned by 6 of the producers, either nest box lights to train the birds, stages of dimmable lights to encourage birds up off the floor during training, or additional lighting in dim floor areas. Five producers also stated they used electric wires on the shed perimeter and/or feeder and drinker lines to prevent birds from perching where they are not meant to, and to prevent laying in the floor area. One producer had the same electric wire layout in rearing, so

the birds learnt early on and were familiar with the wire placement during lay. The electric wires were used predominantly in the training period during early lay. Finally, one producer mentioned the position of the drinker and feeder lines should not be on the litter as it encourages birds to the litter area, and 2 producers stated feeding times should be controlled so there is no conflict between feeding and laying during the morning lay period.

PRODUCER-AIMED ONLINE MANAGEMENT DOCUMENTS

A Google search was conducted December 2022 with the search terms “How to reduce floor egg laying” and “How to prevent floor egg laying.” Websites that were listed across the first 3 to 4 pages of the search were incorporated, with similar themes observed across most of the articles/online management guides that were read. The management suggestions were related to domestic egg laying hens across different cage-free systems, broiler breeder flocks, and backyard hens. There was similarity in management advice across all types of hen systems (detailed in [Supplementary Table 2](#)). Some guidelines were contradictory such as whether to open the nest boxes as soon as the hens arrive in the laying facility or keep them closed for a period to encourage curiosity when the birds come into lay and prevent nest box soiling. Additionally, whether electric wires are beneficial (e.g., prevent laying in floor corners) or detrimental (e.g., discourage hens to jump up if placed on water and feeder lines). As a general summary of the management recommendations, rearing is highlighted as important in training the birds to be able to move up and experience similar conditions to what they would during lay. The training period when the birds first arrive in the laying house is also critical to get the birds up to the nest boxes. Regular floor walks are needed when hens start to come into lay and any eggs on the floor must be picked up immediately. The litter area should not be too comfortable (e.g., litter not too deep, lighting not too dim) and nest boxes should not be uncomfortable (e.g., too draughty, too hot, too bright, infested with mites) or difficult to

access (e.g., hens crowding outside, other housing structures in the way, too far away from feed/water). More information is summarized in [Supplementary Table 2](#), along with source links for each online manual for further details. These factors all align with those documented in a 2018 report based on visits to poultry farms internationally and discussions with poultry experts on management of floor egg laying (O'Flaherty, 2019). The key conclusion from this Nuffield Scholarship activity as well as across the online management documents were that the causes of floor laying are multifactorial but can be sufficiently managed by good stock people (O'Flaherty, 2019). Each flock can be different and thus it is critical to be able to read the birds and adjust management strategies accordingly (O'Flaherty, 2019). It has also been stated in online management guides that despite best practice, the occurrence of and/or extent of floor laying may be “luck,” indicating all the causes of this behavior in a specific flock of birds are not yet known.

CONCLUSIONS AND APPLICATIONS

1. The development of floor egg laying is multifactorial and can be affected by individual preferences as well as the hen strain. Thus, specific strains of hens may be best suited to particular housing system designs.
2. The occurrence of the behavior can vary across flocks in the same housing system which means a good understanding of bird behavior is critical for optimal management of the undesirable behavior.
3. The rearing period is critical for training pullets to travel upwards. Perches, elevated tiers, or rearing like-to-like will improve pullets' physical and behavioral development.
4. The pullets need time to learn the laying housing system before they come into lay, transfer before 16 wk may aid in this.
5. The period following transfer is critical for ensuring the birds are learning to move up off the floor and into the system to access the nest box.
6. Labor investment when the birds first start laying to keep them up off the floor will

prevent floor egg laying from becoming established in the flock.

7. The nest box needs to be the only comfortable, quiet, enclosed place for the hen to lay her eggs.
8. Nest box use can be affected by individual hen preferences as well as social factors in the flock.
9. Placement of feeders and drinkers should not make nest access difficult nor encourage more hens onto the litter, particularly during the morning egg-laying period. Feed times should not coincide directly with oviposition times.
10. New technologies may assist with detecting and managing egg floor laying, but good labor investment early on by knowledgeable staff can result in negligible floor egg percentages.
11. Preventing floor egg laying from developing is critical, as once the behavior is established in a flock there is very little (or nothing) that can be done to change this.

ACKNOWLEDGMENTS

This research was funded by Australian Eggs. Thank you to all the producers who gave up their time to participate in the interviews.

DISCLOSURES

Dana Campbell reports financial support and administrative support were provided by Australian Eggs Ltd.

SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found in the online version at [doi:10.1016/j.japr.2023.100371](https://doi.org/10.1016/j.japr.2023.100371).

REFERENCES

- Abrahamsson, P., and R. Tauson. 1998. Performance and egg quality of laying hens in an aviary system. *JAPR* 7:225–232.
- Ali, A. B. A., D. L. M. Campbell, D. M. Karcher, and J. M. Siegford. 2016. Influence of genetic strain and access

to litter on spatial distribution of 4 strains of laying hens in an aviary system. *Poult. Sci.* 95:2489–2502.

Ali, A. B. A., D. L. M. Campbell, D. M. Karcher, and J. M. Siegford. 2019a. Daytime occupancy of resources and flooring types by 4 laying hen strains in a commercial system. *J. Vet. Behav.* 31:59–66.

Ali, A. B. A., D. L. M. Campbell, and J. M. Siegford. 2020. A risk assessment of health, production, and resource occupancy for 4 laying hen strains across the lay cycle in a commercial-style aviary system. *Poult. Sci.* 99:4672–4684.

Ali, B. A., M. Toscano, and J. M. Siegford. 2019b. Later exposure to perches and nests reduces individual hens' occupancy of vertical space in an aviary and increases force of falls at night. *Poult. Sci.* 98:6251–6262.

Alm, M., H. Wall, L. Holm, A. Wichman, R. Palme, and R. Tauson. 2015. Welfare and performance in layers following temporary exclusion from the litter area on introduction to the layer facility. *Poult. Sci.* 94:565–573.

Appleby, M. C. 1984. Factors affecting floor laying by domestic laying hens: a review. *Worlds Poult. Sci. J.* 40:241–249.

Appleby, M. C., I. J. H. Duncan, and H. E. McRae. 1988b. Perching and floor laying by domestic hens: experimental results and their commercial application. *Br. Poult. Sci.* 29:351–357.

Appleby, M. C., G. S. Hogarth, J. A. Anderson, B. O. Hughes, and C. T. Whittemore. 1988a. Performance of a deep litter system for egg production. *Br. Poult. Sci.* 29:735–751.

Appleby, M. C., G. S. Hogarth, and B. O. Hughes. 1988c. Nest box design and nesting material in a deep litter house for laying hens. *Br. Poult. Sci.* 29:215–222.

Appleby, M. C., S. N. Maguire, and H. E. McRae. 1986. Nesting and floor laying by domestic hens in a commercial flock. *Br. Poult. Sci.* 27:75–82.

Appleby, M. C., and H. E. McRae. 1986. The individual nest box as a super-stimulus for domestic hens. *Appl. Anim. Behav. Sci.* 15:169–176.

Bécot, L., N. Bédère, J. Coton, T. Burlot, and P. Le Roy. 2023. Nest preference and laying duration traits to select against floor eggs in laying hens. *Genet. Sel.* 55:8.

Becot, L., N. Bedere, T. Burlot, J. Coton, and P. Le Roy. 2021. Nest acceptance, clutch, and oviposition traits are promising selection criteria to improve egg production in cage-free system. *PLoS One* 16:e0251037.

Bestman, M., and J.-P. Wagenaar. 2014. Health and welfare in Dutch organic laying hens. *Animals* 4:374–390.

Bist, R. B., X. Yang, S. Subedi, and L. Chai. 2023. Mis-laying behavior detection in cage-free hens with deep learning technologies. *Poult. Sci.* 102:102729.

Brake, J. 1985. Comparison of two nesting materials for broiler breeders. *Poult. Sci.* 64:2263–2266.

Brake, J. 1987. Influence of presence of perches during rearing on incidence of floor laying in broiler breeders. *Poult. Sci.* 66:1587–1589.

Brake, J. 1993. Influence of nest pad color on nest preference, percentage of floor eggs, and egg production of broiler breeder hens. *Poult. Sci.* 72:1663–1666.

Buchwalder, T., and E. K. Fröhlich. 2011. Assessment of colony nests for laying hens in conjunction with the authorization procedure. *Appl. Anim. Behav. Sci.* 134:64–71.

Campbell, D. L. M., E. N. de Haas, and C. Lee. 2019. A review of environmental enrichment for laying hens during rearing in relation to their behavioral and physiological development. *Poult. Sci.* 98:9–28.

Campbell, D. L. M., T. R. Dyall, J. A. Downing, A. M. Cohen-Barnhouse, and C. Lee. 2020. Rearing enrichments affected ranging behavior in free-range laying hens. *Front. Vet. Sci.* 7:446.

Campbell, D. L. M., D. M. Karcher, and J. M. Siegford. 2016. Location tracking of individual laying hens housed in aviaries with different litter substrates. *Appl. Anim. Behav. Sci.* 184:74–79.

Campo, J. L., M. G. Gil, and S. G. Dávila. 2007. Differences among white-, tinted-, and brown-egg laying hens for incidence of eggs laid on the floor and for oviposition time. *Arch. Geflügelk.* 71:S105–S109.

Chang, C.-L., B.-X. Xie, and C.-H. Wang. 2020. Visual guidance and egg collection scheme for a smart poultry robot for free-range farms. *Sensors* 20:6624.

Chien, Y.-R., and Y.-X. Chen. 2018. An RFID-based smart nest box: an experimental study of laying performance and behavior of individual hens. *Sensors* 18:859.

Clausen, T., and A. B. Riber. 2012. Effect of heterogeneity of nest boxes on occurrence of gregarious nesting in laying hens. *Appl. Anim. Behav. Sci.* 142:168–175.

Colson, S., C. Arnould, and V. Michel. 2008. Influence of rearing conditions of pullets on space use and performance of hens placed in aviaries at the beginning of the laying period. *Appl. Anim. Behav. Sci.* 111:286–300.

Cooper, J. J., and M. C. Appleby. 1996. Individual variation in prelaying behaviour and the incidence of floor eggs. *Br. Poult. Sci.* 37:245–253.

Cooper, J. J., and M. C. Appleby. 1997. Motivational aspects of individual variation in response to nestboxes by laying hens. *Anim. Behav.* 54:1245–1253.

Cooper, J. J., and M. J. Albentosa. 2003. Behavioural priorities of laying hens. *Avian Poult. Biol. Rev.* 14:127–149.

Cronin, G. M., J. L. Barnett, and P. H. Hemsworth. 2012. The importance of pre-laying behaviour and nest boxes for laying hen welfare: a review. *Anim. Prod. Sci.* 52:398–405.

Cronin, G. M., K. L. Butler, M. A. Desnoyers, and J. L. Barnett. 2005. The use of nest boxes by hens in cages: what does it mean for welfare? *Anim. Sci. Pap.* 23:121–128.

Donaldson, C. J., and N. E. O'Connell. 2012. The influence of access to aerial perches on fearfulness, social behaviour and production parameters in free-range laying hens. *Appl. Anim. Behav. Sci.* 142:51–60.

Farkas, T. P., S. Szász, A. Orbán, D. Mezőszentgyörgyi, L. Pető, and Z. Sütő. 2022. Examination of nesting behavior of laying hens of different genotypes housed in alternative pens using a video system. *Appl. Sci.* 12:9093.

Freire, R., M. C. Appleby, and B. O. Hughes. 1996. Effects of nest quality and other cues for exploration on pre-laying behaviour. *Appl. Anim. Behav. Sci.* 48:37–46.

Freire, R., M. C. Appleby, and B. O. Hughes. 1997. Assessment of pre-laying motivation in the domestic hen using social interaction. *Anim. Behav.* 54:313–319.

Giersberg, M. F., N. Kemper, and B. Spindler. 2019. Pecking and piling: the behaviour of conventional layer hybrids and dual-purpose hens in the nest. *Appl. Anim. Behav. Sci.* 214:50–56.

- Guinebretière, M., A. Huneau-Salaün, D. Huonnic, and V. Michel. 2012. Cage hygiene, laying location, and egg quality: the effects of linings and litter provision in furnished cages for laying hens. *Poult. Sci.* 91:808–816.
- Gunnarsson, S., L. J. Keeling, and J. Svedberg. 1999. Effect of rearing factors on the prevalence of floor eggs, cloacal cannibalism and feather pecking in commercial flocks of loose housed laying hens. *Br. Poult. Sci.* 40:12–18.
- Heerkens, J. L. T., E. Delezie, B. Ampe, T. B. Rodenburg, and F. A. M. Tuytens. 2016. Ramps and hybrid effects on keel bone and foot pad disorders in modified aviaries for laying hens. *Poult. Sci.* 95:2479–2488.
- Heerkens, J. L. T., E. Delezie, I. Kempen, J. Zoons, B. Ampe, T. B. Rodenburg, and F. A. M. Tuytens. 2015. Specific characteristics of the aviary housing system affect plumage condition, mortality and production in laying hens. *Poult. Sci.* 94:2008–2017.
- Hemsworth, P. H., and L. E. Edwards. 2021. Natural behaviours, their drivers and their implications for laying hen welfare. *Anim. Prod. Sci.* 61:915–930.
- Holcman, A., S. Malovrh, and I. Stuhec. 2007. Choice of nest types by hens of three lines of broiler breeders. *Br. Poult. Sci.* 48:284–290.
- Huber, H. U., D. W. Fölsch, and U. Stähli. 1985. Influence of various nesting materials on nest site selection of the domestic hen. *Br. Poult. Sci.* 26:367–373.
- Huber-Eicher, B. 2004. The effect of early colour preference and of a colour exposing procedure on the choice of nest colours in laying hens. *Appl. Anim. Behav. Sci.* 86:63–76.
- Hughes, B. O. 1993. Choice between artificial turf and wire floor as nest sites in individually caged laying hens. *Appl. Anim. Behav. Sci.* 36:327–335.
- Hunniford, M. E., G. J. Mason, and T. M. Widowski. 2018. Laying hens' preferences for nest surface type are affected by enclosure. *Appl. Anim. Behav. Sci.* 201:7–14.
- Hunniford, M. E., and T. M. Widowski. 2016. Rearing environment and laying location affect pre-laying behaviour in enriched cages. *Appl. Anim. Behav. Sci.* 181:205–213.
- Hunniford, M. E., and T. M. Widowski. 2018. Curtailed nests facilitate settled nesting behaviour of laying hens in furnished cages. *Appl. Anim. Behav. Sci.* 202:39–45.
- Icken, W., D. Cavero, M. Schmutz, and R. Preisinger. 2012. New phenotypes for new breeding goals in layers. *Worlds Poult. Sci. J.* 68:387–400.
- Icken, W., S. Thurner, A. Heinrich, A. Kaiser, D. Cavero, G. Wendl, R. Fries, M. Schmutz, and R. Preisinger. 2013. Higher precision level at individual laying performance tests in noncage housing systems. *Poult. Sci.* 92:2276–2282.
- Janczak, A. M., and A. B. Riber. 2015. Review of rearing-related factors affecting the welfare of laying hens. *Poult. Sci.* 94:1454–1469.
- Joffe, B. P., and C. T. Usher. 2017. Autonomous robotic system for picking up floor eggs in poultry houses. Page 1 in 2017 ASABE Annual International Meeting. American Society of Agricultural and Biological Engineers.
- Jones, D. R., N. A. Cox, J. Guard, P. J. Fedorka-Cray, R. J. Buhr, R. K. Gast, Z. Abdo, L. L. Rigsby, J. R. Plumblee, D. M. Karcher, C. I. Robison, R. A. Blatchford, and M. M. Makagon. 2015. Microbiological impact of three commercial laying hen housing systems. *Poult. Sci.* 94:544–551.
- Kang, H. K., S. B. Park, S. H. Kim, and C. H. Kim. 2016. Effects of stock density on the laying performance, blood parameter, corticosterone, litter quality, gas emission and bone mineral density of laying hens in floor pens. *Poult. Sci.* 95:2764–2770.
- Karcher, D. M., D. R. Jones, C. I. Robison, K. N. Eberle, R. K. Gast, and K. E. Anderson. 2019. Production and well-being resulting from delayed movement of pullets to the hen facility. *JAPR* 28:278–289.
- Krause, E. T., and L. Schrader. 2018. High, low, or familiar? Nest site preferences of experienced laying hens. *Br. Poult. Sci.* 59:359–364.
- Kruschwitz, A., M. Zupan, T. Buchwalder, and B. Huber-Eicher. 2008a. Nest preference of laying hens (*Gallus gallus domesticus*) and their motivation to exert themselves to gain nest access. *Appl. Anim. Behav. Sci.* 112:321–330.
- Kruschwitz, A., M. Zupan, T. Buchwalder, and B. Huber-Eicher. 2008b. Prelaying behaviour of laying hens (*Gallus gallus domesticus*) in different free range settings. *Arch. Geflügelk.* 72:S84–S89.
- Lambton, S. L., T. G. Knowles, C. Yorke, and C. J. Nicol. 2010. The risk factors affecting the development of gentle and severe feather pecking in loose housed laying hens. *Appl. Anim. Behav. Sci.* 123:32–42.
- Lentfer, T. L., S. G. Gebhardt-Henrich, E. K. F. Fröhlich. 2008. Changing nest positions: influence on laying behaviour of laying hens (*Gallus gallus domesticus*) in aviary systems. Pages 125–138 in Association for Technology and Structures in Agriculture (KTBL), Volume 471. M. Erhard, ed. Current Research in Applied Ethology, Darmstadt, Germany eds.[Abstract in English].
- Lentfer, T. L., S. G. Gebhardt-Henrich, E. K. F. Fröhlich, and E. von Borell. 2011. Influence of nest site on the behaviour of laying hens. *Appl. Anim. Behav. Sci.* 135:70–77.
- Lentfer, T. L., S. G. Gebhardt-Henrich, E. K. F. Fröhlich, and E. von Borell. 2013. Nest use is influenced by the positions of nests and drinkers in aviaries. *Poult. Sci.* 92:1433–1442.
- Li, G., G. D. Chesser, Y. Huang, Y. Zhao, and J. L. Purswell. 2021. Development and optimization of a deep-learning-based egg-collecting robot. *Trans. ASABE* 64:1659–1669.
- Li, G., X. Hui, Y. Zhao, W. Zhai, J. L. Purswell, Z. Porter, S. Poudel, L. Jia, B. Zhang, and G. D. Chesser. 2022. Effects of ground robot manipulation on hen floor egg reduction, production performance, stress response, bone quality, and behavior. *PLoS One* 17:e0267568.
- Li, G., Y. Xu, Y. Zhao, Q. Du, and Y. Huang. 2020. Evaluating convolutional neural networks for cage-free floor egg detection. *Sensors* 20:332.
- Li, X., L. Yang, X. Wang, Z. Shi, B. Li, and Q. Tong. 2018. Egg laying patterns of hens in a modified aviary system. ASABE, ILES18-086. Presented at the 10th International Livestock Environment Symposium (ILES X) Nebraska, USA, September 25–27.
- Lillpers, K. 1991. Genetic variation in the time of oviposition in the laying hen. *Br. Poult. Sci.* 32:303–312.
- Lundberg, A., and L. J. Keeling. 1999. The impact of social factors on nesting in laying hens (*Gallus gallus domesticus*). *Appl. Anim. Behav. Sci.* 64:57–69.

- MacLachlan, S. S., A. B. A. Ali, M. J. Toscano, and J. M. Siegford. 2020. Influence of later exposure to perches and nests on flock level distribution of hens in an aviary system during lay. *Poult. Sci.* 99:30–38.
- Mirosh, L. W., J. McGinnis, and W. Sperry. 1986. Environmental factors affecting the egg laying habits of white leghorns. *Poult. Sci.* 65:693–695.
- Nannoni, E., G. Buonaiuto, G. Martelli, G. Lizzi, G. Trevisani, G. Garavini, and L. Sardi. 2022. Influence of increased freedom of movement on welfare and egg laying pattern of hens kept in aviaries. *Animals* 12:2307.
- Nicol, C. J., C. Pötzsch, K. Lewis, and L. E. Green. 2003. Matched concurrent case-control study of risk factors for feather pecking in hens on free-range commercial farms in the UK. *Br. Poult. Sci.* 44:515–523.
- O'Flaherty, E. 2019. Contributing Factors to Floor Egg Issues: *Avoiding the Issues With Best Practices*. Nuffield Australia, Sydney, NSW. https://www.nuffieldscholar.org/sites/default/files/reports/2018_AU_Emma-Ofaherty_Contributing-Factors-To-Floor-Egg-Issues.pdf. Accessed June 2023.
- Oliveira, J. L., H. Xin, L. Chai, and S. T. Millman. 2019. Effects of litter floor access and inclusion of experienced hens in aviary housing on floor eggs, litter condition, air quality, and hen welfare. *Poult. Sci.* 98:1664–1677.
- Palczynski, L. J., H. Buller, S. L. Lambton, and C. A. Weeks. 2016. Farmer attitudes to injurious pecking in laying hens and to potential control strategies. *Anim. Welf.* 25:29–38.
- Pearl, R., and F. M. Surface. 1909. A Biometrical Study of Egg Production in the Domestic Fowl. 1. Variation in Annual Egg Production. (p. 110). U.S. Department of Agriculture Bureau of Annual Industry Bulletin, Washington, D.C., 110.
- Petherick, J. C., E. Seawright, and D. Waddington. 1993. Influence of quantity of litter on nest box selection and nesting behaviour of domestic hens. *Br. Poult. Sci.* 34:847–872.
- Philippe, F. X., Y. Mahmoudi, D. Cinq-Mars, M. Lefrançois, N. Moula, J. Palacios, F. Pelletier, and S. Godbout. 2020. Comparison of egg production, quality and composition in three production systems for laying hens. *Livest. Sci.* 232:103917.
- Pillan, G., G. Xiccato, C. Ciarelli, F. Bordignon, A. Concollato, A. Pascual, M. Birolo, F. Pirrone, F. Sirri, X. Averós, I. Estevez, and A. Trocino. 2023. Factors affecting space use by laying hens in a cage-free aviary system: effect of nest lighting at pullet housing and of curtain nest color during laying. *Poult. Sci.* 102:102524.
- Preisinger, R. 2018. Innovative layer genetics to handle global challenges in egg production. *Br. Poult. Sci.* 59:1–6.
- Purdum, S., P. Eusebio, and K. Hanford. 2020. The effects of 2 genetic lines on spatial distribution and use and preference of perch and nest area in an aviary system. *Poult. Sci.* 99:3328–3333.
- Rana, M. S., C. Lee, J. M. Lea, and D. L. M. Campbell. 2022. Relationship between sunlight and range use of commercial free-range hens in Australia. *PLoS One* 17:e0268854.
- Rayner, A. C., R. Gill, D. Brass, T. H. Willings, and A. Bright. 2016. Smothering in UK free-range flocks. Part 2: investigating correlations between disease, housing and management practices. *Vet. Rec.* 179:252.
- Ren, G., T. Lin, Y. Ying, G. Chowdhary, and K. C. Ting. 2020. Agricultural robotics research applicable to poultry production: a review. *Comput. Electron. Agri.* 169:105218.
- Riber, A. B. 2010. Development with age of nest box use and gregarious nesting in laying hens. *Appl. Anim. Behav. Sci.* 123:24–31.
- Riber, A. B. 2012a. Gregarious nesting: an anti-predator response in laying hens. *Appl. Anim. Behav. Sci.* 138:70–78.
- Riber, A. B. 2012b. Nest sharing under semi-natural conditions in laying hens. *Appl. Anim. Behav. Sci.* 136:44–49.
- Riber, A. B., and D. A. Guzman. 2017. Effects of different types of dark brooders on injurious pecking damage and production-related traits at rear and lay in layers. *Poult. Sci.* 96:3529–3538.
- Riber, A. B., and B. L. Nielsen. 2013. Changes in position and quality of preferred nest box: Effects on nest box use by laying hens. *Appl. Anim. Behav. Sci.* 148:185–191.
- Rietveld-Piepers, B., H. J. Blokhuis, and P. R. Wiepkema. 1985. Egg-laying behaviour and nest-site selection of domestic hens kept in small floor-pens. *Appl. Anim. Behav. Sci.* 14:75–88.
- Ringgenberg, N., E. K. F. Fröhlich, A. Harlander-Matauschek, M. J. Toscano, H. Würbel, and B. A. Roth. 2015a. Nest choice in laying hens: effects of nest partitions and social status. *Appl. Anim. Behav. Sci.* 169:43–50.
- Ringgenberg, N., E. K. F. Fröhlich, A. Harlander-Matauschek, M. J. Toscano, H. Würbel, and B. A. Roth. 2015b. Effects of variation in nest curtain design on pre-laying behaviour of domestic hens. *Appl. Anim. Behav. Sci.* 170:34–43.
- Ringgenberg, N., E. K. F. Fröhlich, A. Harlander-Matauschek, H. Würbel, and B. A. Roth. 2014. Does nest size matter to laying hens? *Appl. Anim. Behav. Sci.* 155:66–73.
- Rizzi, C. 2020. Yield performance, laying behaviour traits and egg quality of purebred and hybrid hens reared under outdoor conditions. *Animals* 10:584.
- Schreiter, R., K. Damme, M. Klunker, C. Raoult, E. von Borell, and M. Freick. 2020. Effects of edible environmental enrichments during the rearing and laying periods in a littered aviary—Part 2: physical development of pullets and performance, egg quality, and carcass composition in laying hens. *Poult. Sci.* 99:6685–6696.
- Settar, P., J. A. Arango, N. P. O'Sullivan, and J. A. Arthur. 2006. Evidence of genetic variability for floor and nest egg laying behavior in floor pens. *Proceedings of the XII European Poultry Conference: 10–14 September 2006; Verona*. Accessed Mar. 2023. <https://www.cabi.org/Uploads/animal-science/worlds-poultry-science-association/WPSA-italy-2006/10194.pdf>.
- Sherwin, C. M., and C. J. Nicol. 1993a. Factors influencing floor-laying by hens in modified cages. *Appl. Anim. Behav. Sci.* 36:211–222.
- Sherwin, C. M., and C. J. Nicol. 1993b. A descriptive account of the pre-laying behaviour of hens housed individually in modified cages with nests. *Appl. Anim. Behav. Sci.* 38:49–60.
- Sherwin, C. M., and C. J. Nicol. 1994. Dichotomy of choice of nest characteristics by caged laying hens. *Anim. Welf.* 3:313–320.
- Silversides, F. G., R. Singh, K. M. Cheng, and D. R. Korver. 2012. Comparison of bones of 4 strains of laying hens kept in conventional cages and floor pens. *Poult. Sci.* 91:1–7.

- Singh, R., K. M. Cheng, and F. G. Silversides. 2009. Production performance and egg quality of four strains of laying hens kept in conventional cages and floor pens. *Poult. Sci.* 88:256–264.
- Son, J., W.-D. Lee, H.-J. Kim, B.-S. Kang, and H.-K. Kang. 2022. Effect of providing environmental enrichment into aviary house on the welfare of laying hens. *Animals* 12:1165.
- Sørensen, P., T. Okeno, and A. J. Buitenhuis. 2017. The motivation of hens to lay eggs on the floor in non-cage systems has a heritable background. *Eur. Poult. Sci.* 81:1–12.
- Stämpfli, K., T. Buchwalder, E. K. F. Fröhlich, and B. A. Roth. 2012. Influence of front curtain design on nest choice by laying hens. *Br. Poult. Sci.* 53:553–560.
- Stämpfli, K., T. Buchwalder, E. K. F. Fröhlich, and B. A. Roth. 2013. Design of nest access grids and perches in front of the nests: influence on the behavior of laying hens. *Poult. Sci.* 92:890–899.
- Stämpfli, K., B. A. Roth, T. Buchwalder, and E. K. F. Fröhlich. 2011. Influence of nest-floor slope on the nest choice of laying hens. *Appl. Anim. Behav. Sci.* 135:286–292.
- Steenfeldt, S., and B. L. Nielsen. 2015. Welfare of organic laying hens kept at different indoor stocking densities in a multi-tier aviary system. I: egg laying, and use of veranda and outdoor area. *Animal* 9:1509–1517.
- Stratmann, A., E. K. F. Fröhlich, S. G. Gebhardt-Henrich, A. Harlander-Matauschek, H. Würbel, and M. J. Toscano. 2015. Modification of aviary design reduces incidence of falls, collisions and keel bone damage in laying hens. *Appl. Anim. Behav. Sci.* 165:112–123.
- Stratmann, A., E. K. F. Fröhlich, S. G. Gebhardt-Henrich, A. Harlander-Matauschek, H. Würbel, and M. J. Toscano. 2016. Genetic selection to increase bone strength affects prevalence of keel bone damage and egg parameters in commercially housed laying hens. *Poult. Sci.* 95:975–984.
- Struelens, E., F. A. M. Tuytens, A. Janssen, T. Leroy, L. Audoor, E. Vranken, K. de Baere, F. Ödberg, D. Berckmans, J. Zoons, and B. Sonck. 2005. Design of laying nests in furnished cages: influence of nesting material, nest box position and seclusion. *Br. Poult. Sci.* 46:9–15.
- Struelens, E., A. Van Nuffel, F. A. M. Tuytens, L. Audoor, E. Vranken, J. Zoons, D. Berckmans, F. Ödberg, S. Van Dongen, and B. Sonck. 2008. Influence of nest seclusion and nesting material on pre-laying behaviour of laying hens. *Appl. Anim. Behav. Sci.* 112:106–119.
- Subedi, S., R. Bist, X. Yang, and L. Chai. 2023. Tracking floor eggs with machine vision in cage-free hen houses. *Poult. Sci.* 102:102637.
- Sun, H., P. Zhao, Y. Su, Q. Zhao, X. Li, and J. Bao. 2023. Effects of different cage sizes and nest setting on laying hen nesting behavior and the proportion of dirty eggs in furnished cages. *Appl. Anim. Behav. Sci.* 262:105917.
- Tahamtani, F. M., L. K. Hinrichsen, and A. B. Riber. 2018. Laying hens performing gregarious nesting show less pacing behaviour during the pre-laying period. *Appl. Anim. Behav. Sci.* 202:46–52.
- Tauson, R., A. Wahlström, and P. Abrahamsson. 1999. Effect of two floor housing systems and cages on health, production, and fear response in layers. *JAPR* 8:152–159.
- Usher, C. T., W. D. Daley, B. P. Joffe, and A. Muni. 2017. Robotics for poultry house management. *Proc. ASABE* 1701103.
- van den Oever, A., J. E. Bolhuis, L. J. F. van de Ven, B. Kemp, and T. B. Rodenburg. 2020b. High levels of contact dermatitis and decreased mobility in broiler breeders, but neither have a relationship with floor eggs. *Poult. Sci.* 99:3355–3362.
- van den Oever, A. C. M., L. Candelotto, B. Kemp, T. B. Rodenburg, J. E. Bolhuis, E. A. M. Graat, L. J. F. van de Ven, D. Guggisberg, and M. J. Toscano. 2021b. Influence of a raised slatted area in front of the nest on leg health, mating behaviour and floor eggs in broiler breeders. *Animal* 15:100109.
- van den Oever, A. C. M., B. Kemp, T. B. Rodenburg, L. J. F. van de Ven, and J. E. Bolhuis. 2021a. Gregarious nesting in relation to floor eggs in broiler breeders. *Animal* 15:100030.
- van den Oever, A. C. M., T. B. Rodenburg, J. E. Bolhuis, L. J. F. van de Ven, M. K. Hasan, S. M. W. van Aerle, and B. Kemp. 2020a. Relative preference for wooden nests affects nesting behaviour of broiler breeders. *Appl. Anim. Behav. Sci.* 222:104883.
- van Horne, P. L. M. 1996. Production and economic results of commercial flocks with white layers in aviary systems and battery cages. *Br. Poult. Sci.* 37:255–261.
- Villanueva, S., A. B. A. Ali, D. L. M. Campbell, and J. M. Siegford. 2017. Nest use and patterns of egg laying and damage by 4 strains of laying hens in an aviary system. *Poult. Sci.* 96:3011–3020.
- Vroegindewij, B. A., S. K. Blaauw, J. M. M. Jsselmuiden, and E. J. van Henten. 2018. Evaluation of the performance of PoultryBot, an autonomous mobile robotic platform for poultry houses. *Biosyst. Eng.* 174:295–315.
- Vroegindewij, B. A., G. L. van Willigenburg, P. W. G. Groot Koerkamp, and E. J. van Henten. 2014. Path planning for the autonomous collection of eggs on floors. *Biosyst. Eng.* 121:186–199.
- Wall, H., R. Tauson, and K. Elwinger. 2002. Effect of nest design, passages, and hybrid on use of nest and production performance of layers in furnished cages. *Poult. Sci.* 81:333–339.
- Wall, H., R. Tauson, and K. Elwinger. 2004. Pop hole passages and welfare in furnished cages for laying hens. *Br. Poult. Sci.* 45:20–27.
- Weeks, C. A., and C. J. Nicol. 2006. Behavioural needs, priorities and preferences of laying hens. *Worlds Poult. Sci. J.* 62:296–307.
- Wolc, A., P. Settar, J. E. Fulton, J. Arango, K. Rowland, D. Lubritz, and J. C. M. Dekkers. 2021. Heritability of perching behavior and its genetic relationship with incidence of floor eggs in Rhode Island Red chickens. *Genet. Sel.* 53:38.
- Zhao, Y., T. A. Shepherd, J. C. Swanson, J. A. Mench, D. M. Karcher, and H. Xin. 2015. Comparative evaluation of three egg production systems: housing characteristics and management practices. *Poult. Sci.* 94:475–484.
- Zheng, H. Y., B. M. Li, G. Chen, and C. Y. Wang. 2018. Improving utilization of nests and decreasing mislaid eggs with narrow width of group nests. *Int. J. Agric. Biol.* 11:83–87.
- Zupan, M., A. Kruschwitz, T. Buchwalder, B. Huber-Eicher, and I. Stuhc. 2008. Comparison of the prelaying behavior of nest layers and litter layers. *Poult. Sci.* 87:399–404.
- Zupan, M., A. Kruschwitz, and B. Huber-Eicher. 2007. The influence of light intensity during early exposure to colours on the choice of nest colours by laying hens. *Appl. Anim. Behav. Sci.* 105:154–164.